# Epic Games Store Giveaways: Causal Effect on Steam Player Engagement

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# Motivation & Research Question

## Motivating example

- Epic Games Store (EGS) has given away over 400 games since 2018 — its primary differentiation lever against Steam
- Standard theory predicts a zero-price substitute on a rival platform diverts players *away* from Steam
- But in January 2026, New Blood CEO Dave Oshry reported that giving away *Blood West* on EGS caused Steam sales to surge $\sim$200% the same day

## Research question

Does an EGS free giveaway *increase* or *decrease* Steam player engagement for the same game — and how does this evolve over time?

## Two competing mechanisms

**Advertising:** Giveaway generates media buzz $\rightarrow$ players flock to Steam (primary discovery/community layer) $\rightarrow$ positive short-run spillover

**Substitution:** Free copy on Epic gradually diverts playtime away from Steam $\rightarrow$ negative medium-run effect

*Both may operate, but at different horizons.*

# Data

## EGS Giveaway History

- Kaggle dataset (Dongre, 2025): every free game on EGS Dec 2018–2025
- 541 events → 471 unique titles; 70 repeat appearances
- 63 titles given away more than once

## Steam Player Data

- Monthly average concurrent players scraped from SteamCharts for each matched game
- Panel: July 2012 – February 2026
- Outcome: $\ln(\texttt{avg_players} + 1)$

## Final Panel

| | |
|---|---|
| Games in final panel | 446 |
| Game-month observations | 38,642 |
| Median game months of data | 88 |
| Mean avg. players/month | 1,584 |
| (median) | 65 |
| Mean $\ln(\texttt{avg_players} + 1)$ | 4.43 |

*Player counts are heavily right-skewed — 101 games have pre-treatment median $< 10$ players. Log transformation compresses the tail.* **Matching:** Fuzzy string matching + manual review; 8 titles confirmed absent from Steam

# Identification Strategy

**Design: Staggered DiD Event Study**

- Treatment = game $i$'s first EGS giveaway, staggered across 80 calendar months (Dec 2018 – late 2025)
- **Control group:** not-yet-treated games — games awaiting future giveaways serve as controls for currently treated games
- No external never-treated group needed

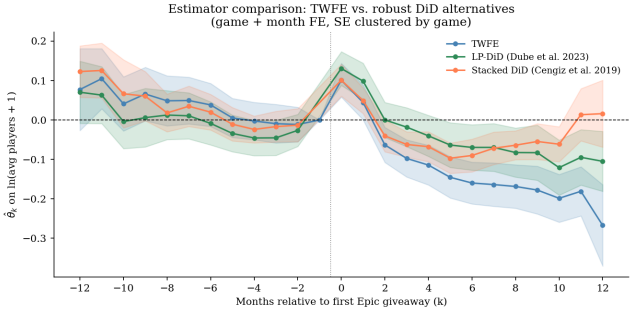
**Key assumption (parallel trends)**
In absence of treatment, log player trajectories would trend identically after conditioning on game & calendar-month FE. Full-sample joint Wald test: $F_{11,444} = 1.72$, $p = 0.067$ (fails to reject at 5%, rejects at 10%). But the low-traffic subsample *does* reject ($p = 0.035$) $\Rightarrow$ lifecycle selection is concentrated in smaller titles.

**Four Estimators (for robustness)**

1. **TWFE** — primary specification; game + calendar-month FE; event window $k \in [-12, 12]$
2. **LP-DiD** (Dubé et al., 2023) — local projections using only not-yet-treated controls at each $k$; robust to heterogeneous effects
3. **Stacked DiD** (Cengiz et al., 2019) — cohort-specific balanced panels; avoids negative-weight bias
4. **CS-ATT** (Callaway & Sant'Anna, 2021) — cohort-specific ATTs; most conservative

**Randomization placebo:** 200 permutations of treatment dates to validate the $k = 0$ spike

# Results: Two-Phase Dynamic




TWFE, LP-DiD, and Stacked DiD. Game + calendar-month FE. $N = 38{,}641$; 446 games. Reference: $k = -1$.

## Phase 1 — Advertising spike

- $+10$ to $+14\%$ more Steam players at $k = 0$ across all estimators
- TWFE and Stacked DiD both $+10.1\%$; LP-DiD $+13.1\%$ (upper bound)
- Placebo $p < 0.001$; actual estimate is $2.7\times$ the 95th percentile of null

## Phase 2 — Medium-run decline

- All estimators turn negative by $k = 2$; significant from $k = 5$ onward
- 0 to $-23\%$ below baseline by $k = 12$ (TWFE most negative; preferred Stacked DiD near zero)
- Partly reflects pre-existing lifecycle decline

# Robustness: Why Do Estimators Differ?

**Control-group composition and calendar-time absorption drive the ordering**

**TWFE** pools all controls and is vulnerable to negative weighting under heterogeneous effects $\Rightarrow$ most negative at large $k$ (bias from pre-existing decline of selected titles).

**LP-DiD** uses *not-yet-treated* controls — games approaching their own giveaway and already in lifecycle decline. Subtracting these declining controls inflates LP-DiD at $k = 0$ and leaves a moderate medium-run negative.

**Stacked DiD** (preferred) adds stack-by-calendar-month FE + treated-interacted event-time dummies (Baker et al. 2022). Most fully absorbs composition and calendar shocks $\Rightarrow$ medium-run negative shrinks toward zero.

| $k$ | TWFE | LP-DiD | Stacked |
|---|---|---|---|
| 0 | $+0.101^{***}$ | $+0.131^{***}$ | $+0.101^{***}$ |
| 1 | $+0.045^{*}$ | $+0.099^{***}$ | $+0.049^{**}$ |
| 2 | $-0.063^{**}$ | 0.000 | $-0.040^{*}$ |
| 5 | $-0.145^{***}$ | $-0.064^{*}$ | $-0.097^{***}$ |
| 10 | $-0.199^{***}$ | $-0.121^{***}$ | $-0.061^{**}$ |
| 12 | $-0.267^{***}$ | $-0.105^{**}$ | 0.016 |

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. *SE clustered by game.*

*All three agree on the $k = 0$ spike and qualitative reversal. Medium-run magnitudes span from $-27\%$ (TWFE) to $+1.6\%$ insignificant (Stacked) — spread bounds true causal substitution.*

# Takeaways, Caveats & Future Work

**Main takeaway:** Advertising spike ($+10$ to $+14\%$ at $k = 0$, robust across all three estimators); medium-run reversal spans $0$ to $-23\%$ across estimators. Preferred Stacked-DiD path is near zero by $k = 12 \Rightarrow$ TWFE/LP-DiD medium-run negatives are best read as upper bounds on causal substitution.

**Developers:** Fee + Steam spike are both positive; player migration to Epic is not a developer cost.

**Steam:** Net positive — spike accrues to Steam; decline is largely pre-existing lifecycle, not loss of engaged users.

**Epic:** Open question — measurable effects accrue to Steam; whether EGS user acquisition offsets the licensing cost is unknown.

## Caveats

- **Lifecycle selection** — pre-trend test rejects on low-traffic half ($p = 0.035$); concern concentrated in smaller titles
- **Repeat null is imprecise** — $\hat{\delta} = 0.0001$ with CI $\pm 12$ log pts
- **Players $\neq$ purchases** — concurrent counts, not sales
- **Genre heterogeneity** — pooled estimate mixes single-player and multiplayer

## Future work

- Single-player vs. multiplayer heterogeneity
- Steam sales volume effects
- EGS-side engagement/sales data